

Xor

$$B \vee C \vee (B \wedge C) \vee (A \wedge B \wedge C)$$

A	B	C	$B \wedge C$	$A \wedge B \wedge C$	expr
T	T	T	T	T	F
T	T	F	F	F	T
T	F	T	F	F	T
T	F	F	F	F	F
F	T	T	T	F	T
F	T	F	F	F	T
F	F	T	F	F	T
F	F	F	F	F	F

$$(\neg A \vee \neg B \vee \neg C) \wedge (B \vee C)$$

A	B	C	$\neg A \vee \neg B \vee \neg C$	$B \vee C$	expr
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	T	T	T
T	F	F	T	F	F
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	F	F

$$\text{len} = 2$$

$$X_{n+2} = a X_{n+1} + b X_n$$

$$\text{len} = 3$$

$$X_{n+3} = a X_{n+2} + b X_{n+1} + c X_n$$

Try $\text{len} = 2$ First

1 1 2 3 2 2 7 9, 1, 4, 29, 24

$$\begin{aligned} \text{len } 2: \quad 2 &= a + b \\ 3 &= 2a + b \end{aligned}$$

$$a = 1$$

$$b = 1$$

$$X_{n+2} = X_{n+1} + X_n$$

$$\text{Check! } X_5 = X_4 + X_3 = 3 + 2 = 5$$

$$X_5 = 2 \neq 5$$

Failed,

$$\text{len } 3: \quad 3 = 2a + b + c \quad a + b = -1$$

$$2 = 3a + 2b + c$$

$$2 = 2a + 3b + 2c$$

$$2a - b = 6 - 2 = 4$$

$$3a = 4 + (-1) = 3$$

$$a = 1$$

$$b = -2$$

$$c = 3$$

$$\begin{aligned} X_{n+3} &= X_{n+2} \\ &\quad - 2X_{n+1} + 3X_n \end{aligned}$$

$$\text{Check! } 7 = 2 - 2 \cdot 2 + 3 \cdot 3 \quad \text{Check rest}$$

$$9 = 7 - 2 \cdot 2 + 3 \cdot 2 \dots$$

Power Set = all subsets

$$P(\{a, b, c\}) = \{ \emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\} \}$$

0	000	$\emptyset$
1	001	$\{a\}$
2	010	$\{b\}$
3	011	$\{b, c\}$
4	100	
5	101	
6	110	
7	111	

x	y	e^x	e^{-x}
-0.5	6.8	$e^{-0.5}$	$e^{0.5}$
-0.25	4.6	$e^{-0.25}$	$e^{0.25} \approx 1.26$
0	5.1	$e^0 = 1$	$e^0 = 1$
0.25	5.7	$e^{0.25}$	$e^{-0.25}$
0.5	4.4	$e^{0.5}$	$e^{-0.5}$

$\uparrow$
 b

$\uparrow$
 A

$$4.4 = a e^{0.5} + b e^{-0.5}$$

$$Ax = b$$

$$x = \begin{pmatrix} a \\ b \end{pmatrix}$$